三菱ガス化学

SAFETY DATA SHEET

MGC

1. Chemical and company identification

Name of chemical (Product name)	**DMAC**

Supplier's company name, address and phone number

Company name	MITSUBISHI GAS CHEMICAL COMPANY, INC.
Address	5-2, Marunouchi 2-chome, Chiyoda-ku, Tokyo 100-8324, Japan
Department in Charge	Basic Chemicals Division I, Basic Chemicals Business Sector
Telephone Number	+81-3-3283-4779
Facsimile Number	+81-3-3214-0930
Email Address	msdsnc@mgc.co.jp
Emergency Telephone Number	+81-3-6890-8677(Verisk 3E)
Accsess Code	335392
Product code	1-12-1100-2

2. Hazards identification

GHS classification

Physical hazards	Flammable liquids	Category 4
Health hazards	Acute toxicity, inhalation	Category 3
	Serious eye damage/eye irritation	Category 2B
	Carcinogenicity	Category 1B
	Reproductive toxicity	Category 1B
	Specific target organ toxicity, single exposure	Category 3 narcotic effects
	Specific target organ toxicity, repeated exposure	Category 1 (liver)
	Specific target organ toxicity, repeated exposure	Category 2 (respiratory system)
Environmental hazards	The product is not classified according to GHS.	

GHS label elements

Pictograms

Signal words Danger

Hazard statement

H227	Combustible liquid.
H320	Causes eye irritation.
H331	Toxic if inhaled.
H332	Harmful if inhaled.
H336	May cause drowsiness or dizziness.
H350	May cause cancer.
H360	May damage fertility or the unborn child.
H372	Causes damage to organs (liver) through prolonged or repeated exposure.
H373	May cause damage to organs (respiratory system) through prolonged or repeated exposure.

Precautionary statement

Prevention

P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P260	Do not breathe mist/vapors.
P264	Wash thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.
P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves/protective clothing/eye protection/face protection.

Response

P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P311	Call a POISON CENTER/doctor.
P337 + P313	If eye irritation persists: Get medical advice/attention.
P370 + P378	In case of fire: Use appropriate media to extinguish.

Storage

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Disposal

P501	Dispose of contents/container in accordance with local/regional/national/international regulations.

Other hazards which do not result in classification None known.

Supplemental information None.

Main symptoms and emergency overview

Main symptoms Direct contact with eyes may cause temporary irritation.

Emergency overview Harmful if inhaled. May cause damage to organs through prolonged or repeated exposure. May cause cancer.

3. Composition/information on ingredients

Substance or mixture Substance

Chemical name or generic name	CAS Number	Gazette notification ENCS no.	Gazette notification ISHL no.	Concentration (%)
N,N-Dimethylacetamide	127-19-5	(2)-723	(2)-723	≥ 99.5

4. First aid measures

If inhaled Remove victim to fresh air and keep at rest in a position comfortable for breathing. Oxygen or artificial respiration if needed. Do not use mouth-to-mouth method if victim inhaled the substance. Induce artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a poison center or doctor/physician.

If on skin Wash off with soap and water. Get medical attention if irritation develops and persists.

If in eyes Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.

If swallowed Rinse mouth. Get medical attention if symptoms occur.

Most important symptoms/effects, acute and delayed May cause drowsiness or dizziness. Headache. Nausea, vomiting. Irritation of eyes. Exposed individuals may experience eye tearing, redness, and discomfort. Jaundice. Prolonged exposure may cause chronic effects.

Protection of first-aid responders IF exposed or concerned: Get medical advice/attention. If you feel unwell, seek medical advice (show the label where possible). Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves. Show this safety data sheet to the doctor in attendance.

Notes to physician Provide general supportive measures and treat symptomatically. Keep victim warm. Keep victim under observation. Symptoms may be delayed.

5. Fire-fighting measures

Extinguishing media Foam. Powder. Powder, carbon dioxide, dry sand, alcohol (foam)

Extinguishing media to avoid Do not use water jet as an extinguisher, as this will spread the fire.

Protection of fire-fighters Wear protective equipment and work from the windward side.

Specific methods For initial fire extinguishing, use powder, carbon dioxide fire extinguisher, dry sand, etc. Use an Alcofoam fire extinguisher for large-scale fires. In case of a surrounding fire, move the container to a safe place immediately. If it cannot be moved, sprinkle water around it to cool it.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Wear appropriate protective equipment and clothing during clean-up. Do not breathe mist/vapors. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ensure adequate ventilation. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.

Environmental precautions Avoid leaking concentrated liquid to rivers.

Methods and materials for containment and cleaning up

Large Spills: Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible. Use a non-combustible material like vermiculite, sand or earth to soak up the product and place into a container for later disposal. Following product recovery, flush area with water. Small Spills: Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal. Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination.

7. Handling and storage

Handling

Technical measures (e.g. Local and general ventilation)

All equipment used when handling the product must be grounded. Use non-sparking tools and explosion-proof equipment. Use only outdoors or in a well-ventilated area.

Safe handling advice

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Do not handle, store or open near an open flame, sources of heat or sources of ignition. Protect material from direct sunlight. Take precautionary measures against static discharges. Do not breathe mist/vapors. Avoid contact with eyes. Avoid prolonged exposure. When using, do not eat, drink or smoke. Pregnant or breastfeeding women must not handle this product. Should be handled in closed systems, if possible. Wash hands thoroughly after handling. Observe good industrial hygiene practices. Use personal protection recommended in Section 8 of the SDS.

Contact avoidance measures

Strong oxidizing agents. For further information, please refer to section 10 of the SDS.

Hygiene measures

Observe any medical surveillance requirements. When using do not smoke. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants.

Storage

Safe storage conditions

Store locked up. Keep away from heat, sparks and open flame. Store in a cool, dry place out of direct sunlight. Keep container tightly closed. Store in a well-ventilated place. Keep in an area equipped with sprinklers. Store away from incompatible materials (see Section 10 of the SDS).

Safe packaging materials

Store in original tightly closed container.

8. Exposure controls/personal protection

Occupational exposure limits

Japan. OELs - JSOH (Japan Society of Occupational Health: Recommendation of Occupational Exposure Limits)

Components	Type	Value
N,N-Dimethylacetamide (CAS 127-19-5)	TWA	36 mg/m3
		10 ppm

US. ACGIH Threshold Limit Values

Components	Type	Value
N,N-Dimethylacetamide (CAS 127-19-5)	TWA	10 ppm

Biological limit values

ACGIH Biological Exposure Indices

Components	Value	Determinant	Specimen	Sampling Time
N,N-Dimethylacetamide (CAS 127-19-5)	30 mg/g	N-Methylacetamide	Creatinine in urine	*

* - For sampling details, please see the source document.

Exposure guidelines

Japan JSOH OELs: Skin designation

N,N-Dimethylacetamide (CAS 127-19-5) — Can be absorbed through the skin.

US ACGIH Threshold Limit Values: Skin designation

N,N-Dimethylacetamide (CAS 127-19-5) — Danger of cutaneous absorption

Engineering measures

Good general ventilation should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Provide eyewash station.

Personal protective equipment

Respiratory protection

If engineering controls do not maintain airborne concentrations below recommended exposure limits (where applicable) or to an acceptable level (in countries where exposure limits have not been established), an approved respirator must be worn. Gas mask for organic gas

Hand protection

Wear protective gloves. Rubber gloves

Eye protection	Wear safety glasses with side shields (or goggles). Protective glasses (goggles, protective face)
Skin and body protection	Protective clothing, safety helmet, safety shoes/rubber boots, rubber apron.

9. Physical and chemical properties

Physical state	Liquid.
Form	Liquid.
Color	Clear water-white
Odor	Ammoniacal. Aromatic.
Melting point/freezing point	-4 °F (-20 °C)
Boiling point, initial boiling point, and boiling range	329 - 330.8 °F (165 - 166 °C)
Combustibility	Flammable
Lower and upper explosion limit / flammability limit	
Flammability limit - lower (%)	1.8 % estimated
Flammability limit - upper (%)	11.5 % estimated
Explosive limit - lower (%)	1.8 % v/v (100˚C)
Explosive limit - upper (%)	11.5 % v/v (160˚C)
Flash point	145.40 °F (63.00 °C) Closed Cup 158.00 °F (70.00 °C) Open Cup
Auto-ignition temperature	914 °F (490 °C)
Decomposition temperature	None known
pH	Not available.
Kinematic viscosity	Not available.
Solubility(ies)	
Solubility (water)	1000 g/l Miscible with water
Solubility (other)	Soluble in: Ether. Ester. Ketone. Aromatic compounds.
Partition coefficient (n-octanol/water) (log value)	log Pow = -0.77 log Pow = -0.796 (25□)
Vapor pressure	0.33 kPa (20˚C) 176 Pa
Density and/or relative density	
Density	Not available.
Relative density	0.94 (20˚C)
Vapor density	3.01
Particle characteristics	Not available.
Other information	
Oxidizing properties	Not oxidizing.

10. Stability and reactivity

Reactivity	It is stable under normal conditions, but when heated it decomposes to produce a very harmful fume.
Chemical stability	Not available.
Possibility of hazardous reactions	It reacts violently with strong oxidants and chlorine-based hydrocarbons, causing a risk of fire and explosion. Reacts with carbon tetrachloride and other halogen compounds in the presence of iron.
Conditions to avoid	Heat, contact with hazardous substances, contact with plastic.
Incompatible materials	Strong oxidizer, iron / halogen compound.
Hazardous decomposition products	Combustion produces toxic gases such as carbon monoxide and nitrogen oxides.

11. Toxicological information

Acute toxicity	Toxic if inhaled.

Components	Species	Test Results
N,N-Dimethylacetamide (CAS 127-19-5)		
Acute		
Dermal		
LD50	Rabbit	2100 - 3600 mg/kg
Inhalation		
Vapor		
LC50	Rat	4.41 mg/l, 4 hours
Mist		
LC50	Rat	2.5 mg/l, 4 Hours
Oral		
LD50	Rat	3000 - 6000 mg/kg

Skin corrosion/irritation Based on available data, the classification criteria are not met.

Serious eye damage/eye irritation Causes eye irritation.

Respiratory or skin sensitization

Respiratory sensitization Due to partial or complete lack of data the classification is not possible.

Skin sensitization Based on available data, the classification criteria are not met.

Germ cell mutagenicity Based on available data, the classification criteria are not met.

Carcinogenicity May cause cancer.

ACGIH Carcinogens

N,N-Dimethylacetamide (CAS 127-19-5) A3 Confirmed animal carcinogen with unknown relevance to humans.

IARC Monographs. Overall Evaluation of Carcinogenicity

N,N-Dimethylacetamide (CAS 127-19-5) 2B Possibly carcinogenic to humans.

Japan Society for Occupational Health: Carcinogen

N,N-Dimethylacetamide (CAS 127-19-5) 2B Possibly carcinogenic to humans.

Reproductive toxicity May damage fertility or the unborn child.

Specific target organ toxicity - single exposure May cause drowsiness or dizziness.

Specific target organ toxicity - repeated exposure Causes damage to organs (liver) through prolonged or repeated exposure. May cause damage to organs (respiratory system) through prolonged or repeated exposure.

Aspiration hazard Due to partial or complete lack of data the classification is not possible.

12. Ecological information

Ecotoxicological data

Components		Species	Test Results
N,N-Dimethylacetamide (CAS 127-19-5)			
Aquatic			
Acute			
Algae	EC50	Algae	> 500 mg/l, 72 hours
Crustacea	LC50	Mid shrimp	966 mg/l, 96 hours
Fish	LC50	Western mosquitofish (Gambusia affinis)	13300 mg/l, 96 hours

Ecotoxicity The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.

Persistence and degradability No data is available on the degradability of this substance.

Bioaccumulation

Bioaccumulative potential

Octanol/water partition coefficient log Kow

N,N-Dimethylacetamide -0.77

Mobility in soil No data available for this product.

Hazardous to the ozone layer No data available.

Other hazardous effects No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.

13. Disposal considerations

Residual waste Dispose of contents / container in accordance with legal regulations such as Waste Disposal Law.

Contaminated packaging After removing the deposits, dispose them. The cleaning waste liquid is subjected to the same treatment as the residual waste.

Local disposal regulations Collect and reclaim or dispose in sealed containers at licensed waste disposal site.

14. Transport information

IATA

UN number	2810
UN proper shipping name	Toxic liquid, organic, n.o.s.
Transport hazard class(es)	
Class	6.1
Subsidiary risk	-
Packing group	II
Environmental hazards	No.
ERG Code	6L
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Other information	
Passenger and cargo aircraft	Allowed with restrictions.
Cargo aircraft only	Allowed with restrictions.

IMDG

UN number	2810
UN proper shipping name	TOXIC LIQUID, ORGANIC, N.O.S.
Transport hazard class(es)	
Class	6.1
Subsidiary risk	-
Packing group	II
Environmental hazards	
Marine pollutant	No.
EmS	F-A, S-A
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code Not established.

General information Follow regulation in section 15 for domestic transportation.

IATA; IMDG

National regulations Follow regulation in section 15 for domestic transportation.

Emergency Response Guide Number 153

15. Regulatory information

Industrial Safety and Health Act

Notifiable substances

N,N-dimethylacetamide — Table 9 Ordinance No. 284

Labeling substances

N,N-dimethylacetamide

Confirmed carcinogenic chemical substances

N,N-dimethylacetamide

Poisonous and Deleterious Substances Control Act

Specified poisonous substances

Not regulated.

Poisonous substances

Not regulated.

Deleterious substances

Not regulated.

Act on the Regulation of Manufacture and Evaluation of Chemical Substances

Class I specified chemical substances

Not regulated.

Class II specified chemical substances

Not regulated.

Monitoring chemical substances

Not regulated.

Priority Assessment Chemical Substances (PACs)

Not regulated.

Reporting Exempted Substances

Not regulated.

Law concerning Pollutant Release and Transfer Register

Specified class 1 substances (substance name, ordinance number and content)

Not regulated.

Class 1 substances (substance name, ordinance number and content)

N,N-dimethylacetamide Ordinance No. 213 (N,N-Dimethylacetamide)

Class 2 substances (substance name, ordinance number and content)

Not regulated.

Fire Service Act Class 4 Group 2 oils (Water soluble) Hazard rank III (Storage limit: 2000 l)

Ship Safety Law, Dangerous Goods Marine Transport and Storage Rule Toxic substances

Air Law, Enforcement Rule Toxic substances

Explosives Control Act

Not regulated.

Act on Prevention of Marine Pollution and Maritime Disaster

N,N-Dimethylacetamide Category: Z

Air Pollution Control Act

N,N-Dimethylacetoamide

16. Other information

Bibliography

ACGIH Documentation of the Threshold Limit Values and Biological Exposure Indices
HSDB® - Hazardous Substances Data Bank
IARC Monographs. Overall Evaluation of Carcinogenicity
Japan Chemical Industry Association (JCIA) GHS Guideline, June 2019
Japan Society for Occupational Health, Recommendation of Occupational Exposure Limits
JIS Z 7253:2019 Classification of chemicals based on “Globally Harmonized System of Classification and Labelling of Chemicals (GHS)”
JIS Z 7253:2019 Hazard communication of chemicals based on GHS - Labelling and Safety Data Sheet (SDS)
National Toxicology Program (NTP) Report on Carcinogens

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